An attractor account of exact numerosity perception

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Abstract

Numerical cognition is classically divided between two systems: an exact mechanism for small sets and an approximate number system for larger ones, separated by a boundary that is described rather than explained. We propose a mechanistic alternative in which the integers act like memories on the number line: each number is an attractor well whose width grows with magnitude, and noisy sensory evidence is resolved into a discrete percept by pattern-completion dynamics. Fitted at the level of individual responses to brief-exposure estimation data, the model outperforms accumulator-based and Weber-style alternatives; the dynamics improve the fit at equal parameter count and generate response times from convergence time and residual readout uncertainty. The model predicts its own limit: the numerosity at which neighbouring wells stop being resolvable, which grows with exposure and coincides with the response-time plateau commonly found in numerosity estimation. We suggest that a single core number system — the full set of integer attractors — is read in two modes: an exact mode, in which the dynamics resolve individual wells (the Exact Number System, the focus of this work), and an approximate mode, engaged once exact resolution is given up beyond this limit (the Approximate Number System, which we do not model here).

Keywords

numerical cognition, subitizing, exact number system, attractor network, approximate number system

Introduction

Humans perceive small numerosities (1–4) quickly and accurately, a phenomenon known as subitizing (Kaufman et al. 1949). Beyond this range, response time increases and accuracy deteriorates, with variability that grows with set size in the ratio-dependent manner characteristic of Weber's law (Mandler and Shebo 1982; Dehaene 2011). These two signatures — fast exactness for small sets, graded approximation for larger ones — have been documented across ages, cultures, and species (Feigenson et al. 2004; Pica et al. 2004; Nieder 2016), and they anchor the standard view of numerical cognition as the product of two core systems: an Object Tracking System (OTS) that individuates a handful of items exactly, and an Approximate Number System (ANS) that represents larger magnitudes noisily (Piazza et al. 2011; Feigenson et al. 2002; Halberda et al. 2008).

What the two-systems picture does not provide is a mechanistic account of the boundary between them. The limit of exact perception is treated as a property of the OTS — a fixed capacity of three or four items — yet empirically it is not fixed: the range of near-exact responding expands with viewing time (Cheyette and Piantadosi 2020) and contracts in developmental dyscalculia (Decarli et al. 2020). A boundary that moves with exposure and experience is difficult to reconcile with a hand-off between two independent modules, and no current account derives it from the structure of the underlying representation: it simply marks where one explanation stops and another begins.

Computational work has shown that the two regimes need not reflect two mechanisms. Cheyette and Piantadosi (2020) offer the clearest normative formulation: under a power-law prior and a finite information-processing constraint, near-exact performance for small numerosities and Weber-like variability for larger ones emerge from a single optimization. Cheyette et al. (2024) later pushed the argument upstream, proposing that the psychophysics of number follows from bandwidth limits on visuospatial memory rather than from number-specific representations, and Malo et al. (2023) showed that a small quantum spin model can encode numerosity in the spectrum of its collective dynamics. Recurrent networks with biologically grounded on-centre off-surround dynamics likewise produce a partition of the numerosity range from a single architecture (Sengupta et al. 2014), and image-computable networks account for how numerosity is extracted from the visual scene in the first place (Stoianov and Zorzi 2012; Testolin et al. 2020). These accounts establish that unification is possible at the computational level; what is still missing is a mechanistic model of the exact regime itself — one that says what kind of representation makes small numbers exact, and that derives, rather than assumes, where exactness ends.

Here we propose such a model. The mental number line is cast as a one-dimensional attractor landscape in the sense of modern Hopfield networks (Hopfield 1982; Ramsauer et al. 2021): each integer is a stored memory —

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a well whose width grows super-linearly with magnitude, parameterized by the frequency with which numbers occur in the environment (Dehaene and Mehler 1992) — and numerosity judgement is pattern completion, a noisy sensory estimate relaxing to the nearest resolvable well. We evaluate the model on ∼53,000 individual responses from the brief-exposure dataset of Cheyette and Piantadosi (2020), where it outperforms accumulator-style, Weber/ANS, and normative alternatives; its clean-up dynamics improve the fit at equal parameter count and, together with the residual readout uncertainty, generate response times. Crucially, the same landscape predicts its own limit: the numerosity n_{crit} at which neighbouring wells stop being resolvable, which grows with exposure duration and coincides with the onset of the empirical response-time plateau. On this view, a single representation — the full lattice of integer attractors, which we take to be the core number system — is read in two modes. When the wells are resolvable, pattern completion recovers the exact numerosity: this exact mode is the Exact Number System (ENS), and is what the present model characterizes. Beyond n_{crit} the wells have merged and exact resolution is no longer reliable, so responding falls back on an approximate readout of the same landscape — the Approximate Number System (ANS) — whose engagement we do not attempt to model mechanistically here. The transition between the two is therefore not a hand-off between independent modules but one representation reaching its resolution limit; the classical subitizing (Object Tracking System) range corresponds to the fully resolved lower part of the exact mode, so the proposal refines rather than replaces the core-systems picture (Feigenson et al. 2004). As an exploratory extension, we apply the model to exact-enumeration data from children with developmental dyscalculia (Decarli et al. 2020), where it localizes the deficit as a shorter exact range (Supplementary Material 5).

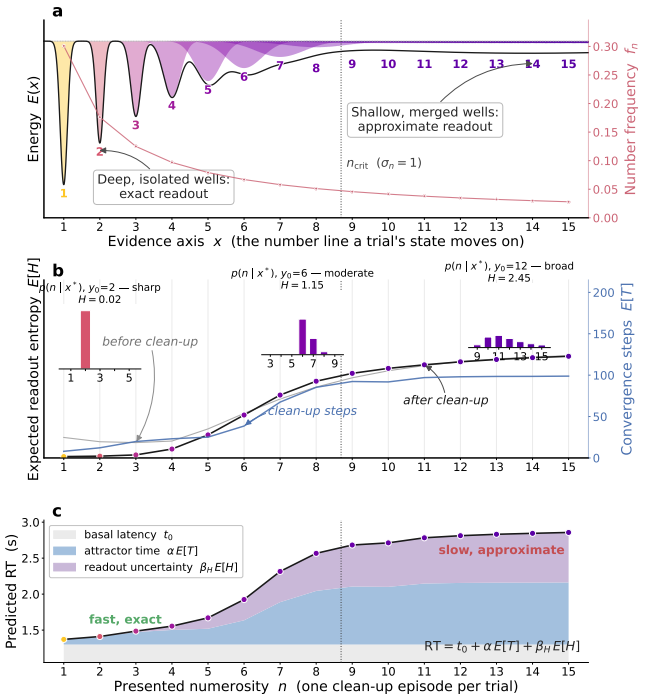


Figure 1. The attractor model of the Exact Number System (parameters fitted at 160 ms). a, Energy landscape over the internal evidence axis x: each integer is a well whose width grows with magnitude (well widths set by number frequency f_n, pink curve), so small numbers sit in deep, isolated basins that clean up to an exact value, whereas beyond n_{crit} (dotted line, where adjacent wells reach $\sigma_n = 1$) the wells merge into a shallow, approximate code. **b**, Expected readout entropy $E[H]$ per presented numerosity, before (grey) and after (black) clean-up, with the attractor's expected convergence time $E[T]$ (blue, right axis); insets show example post-clean-up posteriors. Clean-up removes the residual uncertainty of the small-n readout at the cost of a few settling steps; near the boundary the settling time grows and the residual entropy rises. **c**, The resulting response time, $\text{RT} = t_0 + \tau\, E[T] + \kappa\, E[H]$, decomposed into its basal, convergence-time, and readout-uncertainty components.

Methods

The model maps a noisy perceptual estimate of numerosity onto a discrete response through relaxation on an energy landscape whose stable states sit at the integers (Fig. 1a). A single trial has three stages: a sensory estimate provides a starting point, gradient descent cleans it up toward a well, and a probability distribution over integers is read out from the resting state. The same dynamics also determine how long the trial takes.

Attractor landscape. Each integer n is a stored memory represented by a Gaussian well centred at n, with a width that grows with magnitude. We tie the width to how often each number occurs in the environment: number-word frequency follows an approximately power-law (Benford-like) decline across languages (Dehaene and Mehler 1992), so frequent small numbers occupy narrow, stable basins and rare large numbers occupy broad ones. Writing $f_n = \log_{10}(1 + 1/n)$ for the frequency of n, the well width is

$$\sigma_n = (\sigma_c / f_n)^{\beta}\,, \tag{1}$$

where σ_c sets the overall scale and the exponent β controls how strongly frequency modulates width. We leave β free because there is no a priori reason for the mapping from frequency to width to be linear; as we show below, the data require it to be super-linear ($\beta \approx 2$), which is what makes small numbers sharply resolved and large ones broadly overlapping. Each well contributes a log-likelihood

$$g_n(x) = -\log \sigma_n - \frac{(x-n)^2}{2\,\sigma_n^2}, \tag{2}$$

and, following modern Hopfield networks (Ramsauer et al. 2021), the attraction potential is the log-sum-exp over all wells, $V(x) = -\log \sum_n \exp\big(g_n(x)\big)$.

Clean-up and readout. A sensory estimate y_0 initialises the state; we model the sensory channel as $y_0 \sim \mathcal{N}\big(n,\ s_{00}^2 + s_0\, n^{\lambda_0}\big)$, a neutral front end whose noise grows with numerosity (s_{00} fixed, s_0 and λ_0 fitted). From $x = y_0$ the state descends the potential, $x_{t+1} = x_t - \eta\, \partial V/\partial x$, until it settles or begins to oscillate, giving a resting state x^* reached in T steps. A distribution over integers is then read out, $p(n \mid x^*) = \text{softmax}\big(g_n(x^*)\big)$, from which we take the mean estimate and the most probable integer (exact accuracy).

Response time. The chronometric prediction combines the two quantities the mechanism itself produces: the time

the state takes to settle and the uncertainty that remains once it has,

$$\mathrm{RT} = t_0 + \tau\,\mathbb{E}[T] + \kappa\,\mathbb{E}\big[H(p(n \mid x^*))\big], \qquad (3)$$

where T is the number of clean-up steps, $H = -\sum_n p \log p$ is the residual readout entropy, and expectations are taken over the sensory noise. The convergence-time term $\tau\,\mathbb{E}[T]$ captures the deliberation the attractor dynamics require — long near the boundary, where the landscape flattens and the state settles slowly — while $\kappa\,\mathbb{E}[H]$ captures how undecided the readout remains between neighbouring integers. We compare this hybrid link against its two nested special cases (entropy only, $\tau = 0$; steps only, $\kappa = 0$). The basal latency t_0 is task-specific and fixed per dataset.

Trial-level likelihood. We fit the model to the individual responses of Cheyette and Piantadosi (2020), in whom adult participants ($N = 100$ per condition) briefly viewed dot arrays of 1–15 items and typed an estimate, at five exposure durations (40, 80, 160, 320, 640 ms). Rather than fitting summary curves, we evaluate the categorical likelihood of the ∼53,000 individual integer reports. For each numerosity the response probability marginalises the readout over the sensory noise,

$$p(r \mid n) = \int p\big(r \mid x^*(y_0)\big)\,\phi\big(y_0; n, \sigma(n)\big)\,dy_0, \qquad (4)$$

computed by deterministic grid quadrature (Gauss–Hermite fails on the near-step clean-up integrand; see Supplementary Material 1). Typed responses run well beyond the well support, so we add a uniform lapse channel, $p_{\mathrm{final}}(r \mid n) = (1-\ell)\,p(r \mid n) + \ell\,/\,R_{\max}$, with ℓ fitted per duration and counted in all information criteria. The response model therefore has four mechanism parameters (σ_c, s_0, λ_0, β) plus the lapse, fitted independently per duration by differential evolution (Storn and Price 1997) and Nelder–Mead polishing; the RT channel adds τ and κ. The response, RT, and coupled joint likelihoods are optimised as described in Supplementary Material 1.

Model comparison. To separate the contribution of the attractor dynamics from that of the well shape, we compare the model against a graded set of alternatives fitted under the identical trial-level convention: static (no-clean-up) readouts of the same wells; the same wells with a fixed $\sqrt{n}$ width ($\beta = 1/2$), with and without clean-up; a Weber/ANS code with magnitude-proportional noise; and a log-prior normative proxy in the spirit of Cheyette and Piantadosi (2020). Because turning clean-up on or off leaves the parameter count unchanged, the attractor–vs–static contrast is a zero-parameter comparison. Model quality is summarised by AIC and BIC over the full response count table.

Resolution limit. Finally, the landscape yields a parameter-free prediction of where the exact code breaks down. Adjacent wells stop being reliably distinguishable when their separation falls to one standard deviation, $d' = \mathrm{spacing}/\sigma_n = 1$ (Macmillan and Creelman 2004); solving $\sigma_n = 1$ gives the critical numerosity n_{crit} at which the discrete code loses resolution. This quantity is derived from the fitted landscape, not from any boundary in the data.

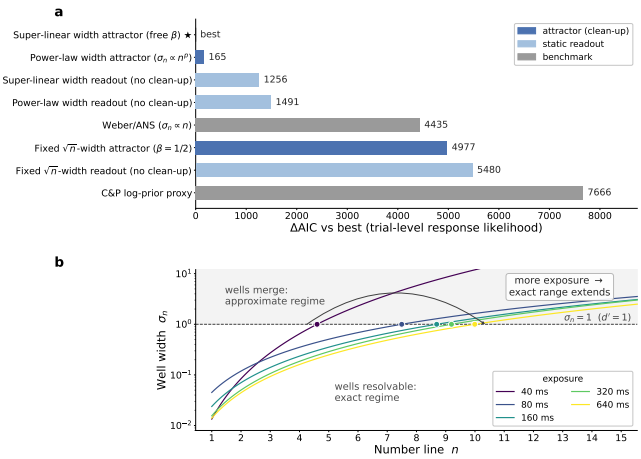


Figure 2. Trial-level model comparison and the predicted resolution limit. a, Response-likelihood AIC relative to the best model, over the ∼53,000 individual responses. The super-linear-width attractor (star) wins; each alternative isolates one ingredient — clean-up (attractor vs. static readout at equal parameters), width law (super-linear vs. fixed $\sqrt{n}$), and the Weber/ANS and normative benchmarks. **b**, The fitted width law σ_n per exposure duration. Wells are resolvable ($\sigma_n < 1$) at small n and merge ($\sigma_n \geq 1$) beyond the critical numerosity n_{crit} (dots), which grows from ≈4.6 at 40 ms to ≈10 at 640 ms: more exposure sharpens the wells and extends the exact range.

Results

The attractor dynamics and the well shape both do real work. At the trial level the models separate sharply (Fig. 2a; full ladder in Supplementary Material 2). The super-linear-width attractor is the best account, ahead of the published accumulator $\sqrt{n}$ model by $\Delta\mathrm{AIC} \approx 4800$ and the Weber/ANS code by ≈ 4400; the normative proxy is furthest behind. Three separable ingredients each carry part of this margin. First, the *well shape*: symmetric Gaussian wells beat the right-skewed accumulator form by $\approx$ 3100–4300 AIC at matched parameter count — human estimates are underdispersed relative to the accumulator's skew, which is the wrong sign for the data. Second, the *width law*: freeing the width exponent, the data demand a super-linear growth ($\beta \approx 2$) rather than the accumulator's $\sqrt{n}$, worth a further $\approx$ 4200–5000 AIC within the Gaussian family. Third, the *clean-up dynamics*: because turning clean-up on is a zero-parameter change, its effect is the cleanest possible test, and it improves AIC by ≈ 1260 (a matched static readout of the same wells is that much worse). The clean-up advantage is concentrated at the briefest exposures, where the sensory evidence is poorest; at 320–640 ms a static readout of the same landscape does about as well on the response distribution, as expected if the attractor has largely settled during the exposure itself, leaving little for the post-stimulus dynamics to resolve.

One landscape fits estimation, accuracy, and response time. With these ingredients the model reproduces all three behavioural signatures jointly (Fig. 3): pooled across durations, $R^2 = 0.99$ for the mean estimate, 0.94 for exact accuracy, and 0.88 for response time, the last a genuine cross-prediction of a channel fitted to the individual trial RTs rather than to a summary curve. The one systematic miss is a mild under-prediction of exact accuracy in the subitizing

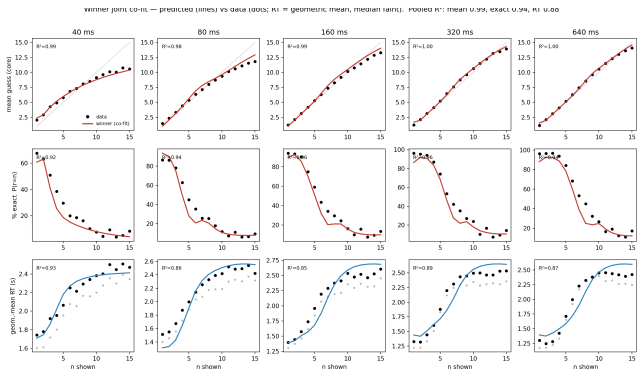


Figure 3. Joint fit of the model to the Cheyette and Piantadosi (2020) data. Rows show presentation times from 40 to 640 ms; columns show the mean estimate, exact accuracy $P(r = n)$, and geometric-mean response time. Dots are empirical data (faint dots in the RT column are the median, shown for reference); lines are the model, co-fitted across all three channels. Response time is a cross-prediction: the RT channel is fitted to the individual trial times, not to the curve shown. R^2 values are given per panel.

band ($n = 2$–4): the model captures the trend toward near-perfect small-number performance but not its full sharpness, an honest residual we return to in the Discussion.

Response time follows from the attractor dynamics. A long-standing objection to attractor accounts is that response time is usually grafted on through a static uncertainty proxy rather than derived from the dynamics. Here the convergence time is available directly, and it matters: the hybrid link that combines settling time and residual entropy (Eq. 3) outperforms an entropy-only link at every one of the five durations (joint $\Delta\mathrm{AIC} \approx 15$–$300$ per duration, $\approx$780 summed), even after paying for its extra parameter. The two terms play complementary roles (Fig. 1b–c): near the boundary the landscape flattens and the state settles slowly, so the convergence-time term drives most of the rise in RT, while the residual-entropy term dominates once the wells have merged. The chronometric prediction is thus not an add-on but a second readout of the same mechanism.

The model predicts its own resolution limit. Because the well width grows with numerosity, the landscape specifies where the discrete code must fail: the numerosity n_{crit} at which adjacent wells reach $\sigma_n = 1$ (Fig. 2b). This crossing falls *inside* the fitted range, at $n_{\mathrm{crit}} \approx 4.6, 7.5, 8.7, 9.2, 10.0$ for the five durations — it is not imposed by any cutoff, since the model is fitted on the full range. Two independent facts converge on it. First, n_{crit} grows monotonically with exposure: more viewing time sharpens the wells, so the exact range extends from $\sim$5 at a 40 ms flash to $\sim$10 at 640 ms. This duration dependence is a property of the landscape, not of a fixed capacity: a constrained fit sharing the landscape parameters across durations is reported as a backstop in Supplementary Material 2, but the substantive point is that the exposure-driven sharpening is exactly what a trainable memory should show. Second, the response-time plateau in the data begins near n_{crit}: within the exact range the model tracks RT closely, and above it the deliberation-time prediction keeps rising while the empirical RT flattens (Fig. 1c and Supplementary Material 4) — the point at which participants stop resolving and start approximating. Accuracy and chronometry, two independent channels, thus mark the same boundary.

A parameter-recovery analysis (Supplementary Material 3) confirms that the quantities we interpret are identifiable at the trial level: the width exponent β and, especially, the derived n_{crit} recover almost perfectly ($R^2 = 0.89$ and 0.996), while the two sensory parameters trade off and are interpreted only through the combined noise profile $\sigma(n)$. A brief exploratory application to developmental dyscalculia, where the same model localises the deficit as a shorter exact range, is reported in Supplementary Material 5.

Discussion

We have described exact numerosity perception as pattern completion on an attractor landscape: the integers are memories whose widths grow with magnitude, and a noisy estimate relaxes to the nearest resolvable well. Fitted to individual responses, the model reproduces estimation, accuracy, and response time together, and — unlike accounts that stipulate a subitizing capacity — it derives the limit of exact perception from the structure of the representation itself. The exact and approximate regimes become two readings of one landscape rather than two systems joined at an unexplained seam.

Two features of the account do the explanatory work, and both survive the trial-level test that a coarser comparison would have hidden. The first is that the dynamics are not decorative. Turning clean-up on is a zero-parameter change, yet it improves the fit substantially, and — more tellingly — the attractor's convergence time predicts response time better than the static uncertainty proxy that such models usually rely on. What makes this possible is the metric, one-dimensional structure of the number line: unlike a generic associative memory storing arbitrary patterns, here the wells lie in a fixed order with unit spacing, so the log-sum-exp coupling of neighbouring wells produces a graded loss of resolution — and a settling time that lengthens toward the boundary — as an intrinsic property of the geometry rather than an added assumption. The second feature is that the well widths are set by number frequency. This ties the representation to the broader view of numbers as memory traces whose strength reflects experience (Anderson and Schooler 1991; Verguts and Fias 2004): frequent small numbers are sharply encoded, rare large ones diffusely.

Casting the code as a frequency-shaped memory has a developmental consequence that a fixed-capacity account lacks. The resolution limit n_{crit} is not a constant but a property of how sharp the wells are, and we already see it move with exposure over milliseconds; over development it should move with numerical experience. As children encounter number words and counting, the wells that carry the most exposure should sharpen, extending the exact range and pushing the transition to approximate perception upward. This predicts that the boundary sits at very small numbers early in development and grows with numeracy, and — more specifically — that it should be trained by exposure to *numbers*, not by general improvement in visual acuity: an intervention that sharpens the number code should differ in its signature from one that improves perception at

large (Butterworth and Laurillard 2017). The developmental dyscalculia analysis (Supplementary Material 5) is an early step in this direction: the same model localises the deficit not as noisier perception but as a shorter exact range — a code whose resolution runs out earlier.

The approximate regime, on this picture, is the same lattice read once exact resolution is abandoned. Beyond n_{crit} the wells have merged, clean-up no longer sharpens the posterior, and the system reads a coarse, ratio-dependent magnitude — the behaviour attributed to the Approximate Number System. We do not model the transition itself: what decides that a trial should stop resolving and report an approximate value is left open, and characterising that switch — and whether phenomena such as numerosity adaptation live in it — is a natural next step. The claim here is only the more limited one that a single representation can support both readings, so the two regimes need not be two mechanisms.

This account offers a different level of explanation from the leading unifying alternative. Cheyette et al. (2024) derive number psychophysics from a bandwidth limit on visuospatial memory; in our model that limit would live in the sensory channel $\sigma(n)$, upstream of the attractor, and the observation that n_{crit} grows with exposure is consistent with more available bandwidth yielding a finer code. The accounts are therefore complementary rather than rival, and we note that both invoke a frequency prior of the Benford family. Where they differ is in what the number-specific representation contributes: our attractor is modality-general (any channel delivering a scalar estimate feeds the same wells) and, being a memory, is intrinsically trainable. It also differs from recurrent accounts that partition the numerosity range through fixed network dynamics (Sengupta et al. 2014): our boundary is not a fixed partition but one that moves with exposure and, we predict, with development — a discriminating test between the two.

Several limitations remain. The evidence channel is deliberately neutral: it summarises the visual front end as a noise profile $\sigma(n)$, so the model operates on an already-extracted estimate. This isolates the attractor dynamics but, by construction, sets aside the influence of continuous magnitudes (area, density, convex hull) on numerosity judgements; coupling the wells to an image-computable front end that is itself sensitive to number (Stoianov and Zorzi 2012; Testolin et al. 2020), and that has been used to simulate dyscalculic profiles (Dolfi et al. 2024), is the clearest way to close this gap. The model also under-predicts the sharpness of small-number accuracy at $n = 2$–4: it captures the trend but not the full abruptness of the subitizing edge, which may still require a fast parallel individuation process for the very smallest sets (Feigenson et al. 2002). The precise form of the effective Weber fraction the model predicts across the estimation range, and how it compares with reanalyses of classical estimation data (Testolin and McClelland 2021), remains to be characterised. Finally, the model is specified at a computational and algorithmic level; relating the wells to the number-tuned populations of parietal cortex, whose tuning widths grow with preferred numerosity (Piazza et al. 2004; Harvey et al. 2013), is a natural direction for testing it neurally.

Numerical cognition, in this framework, is pattern completion on a landscape of number memories. A single set of wells — read exactly where they are resolved and approximately where they merge — accounts for the fast exactness of small numbers, the graded approximation of larger ones, the response times of both, and the moving boundary between them, offering a compact and mechanistic set of hypotheses about how the mind represents number.

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Author Contributions

Eudald Correig-Fraga: Conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing - original draft, and writing - review and editing.

Ethical Considerations

This article reports computational modelling and secondary analysis of previously published, publicly available datasets. No new human participants were recruited, and no new human data were collected for this study. Ethical approval for the original data collections was reported in the source publications.

Consent to Participate

Not applicable. This study did not recruit participants or collect new participant data.

Consent for Publication

Not applicable. This study does not contain identifiable individual participant information.

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Declaration of conflicting interests

The author declared no potential conflicts of interest.

Data Availability

Code is available at https://github.com/eudaldc/hopfield_numbers. The source datasets are available at https://osf.io/svcy5/ and https://osf.io/gjpb3/.

Supplemental material

Supplementary material is available with this article.

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